

The Nexus Of Foreign Reserves, Exchange Rate And Inflation: Recent Empirical Evidence From Sri Lanka

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journals.sagepub.com/home/sae**Ayesh Ariyasinghe¹**  **and N. S. Cooray²**

Abstract

The conventional mandates of the central banks on meeting stability objectives and maintaining a growth-maximizing inflation rate have come under some criticism since the global financial crises. Maintaining adequate foreign reserves is seen as a viable solution to foreign exchange liquidity needs during crisis periods. Since the end of 2011, many Asian economies, including China and Japan, led from the forefront in central bank-led reserves build-up. However, reserves build-up remains challenging and sensitive for small open economies. Such policies help create 'risk-neutral' buffers for monetary and fiscal authorities to absorb transitory current account shocks and foreign exchange stress to smoothen the balance of payments. This study is motivated by the importance of identifying the inflation–foreign reserves nexus that may affect inflation in a manner counter-productive to the central bank mandate of maintaining price stability. It probes the debate of the sustainability of reserves build-up in the long and the short term. The outcome of the study poses several vital questions for fiscal and monetary policymakers concerning their respective mandates. The reserves–inflation nexus and its magnitude is determined using monthly data spanning two decades, through engaging an autoregressive distributed lag (ARDL) model and relevant bounds-testing techniques proposed by Pesaran et al. The vector autoregression (VAR), error correction and Johansen cointegration methods supplement the robustness checks. Exchange rate is introduced to enrich the discussion on the reserves–inflation nexus and shows a cointegration relationship in the long run. The study provides an insight into the influence of exchange rate on reserves and inflation. The variance decomposition shows the presence of a lukewarm response from foreign reserves and exchange rate on inflation. Policymakers concerned with inflationary expectations in the medium-to-long term need to

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consider these signals, as reserves build-up is one of the important policy-driven objectives for a number of economies.

JEL: C50, C32, E31, E52, E58, F30, F31, F39

Keywords

Sri Lanka, foreign reserves, inflation, exchange rate, central bank and monetary policy, ARDL estimation

Introduction

Modern mandates of central banking largely reflect maintaining price stability as having a greater co-dependence on financial system stability (Dow, 2017). Global paradigm shifts highlighting the importance of preserving foreign reserves buffers have added ancillary responsibilities to a central bank's traditional economic stability role. Apart from reserves acting as means to stabilize the exchange rate, adequate foreign reserves tend to smoothen international debt-servicing payment schedules, add to the collateral strength of a sovereign and act as a signal for the economy's welfare. Sovereign reserves build-up are motivated by precautionary and non-precautionary measures (Chițu et al., 2019). Since the end of 2011, many Asian economies, including the two largest economies of China's and Japan's were at the forefront of central bank-led reserves build-up. However, reserves build-up remains challenging for many small open economies, where the masses' welfare requirements may act counter to foreign reserves build-up in the face of the argument that resources could be better utilized for public welfare measures. Sri Lanka's case is no exception to this.

Fiscal and current account twin deficits burden Sri Lanka. The design and implementation of policies oriented towards growth and stability are increasingly challenging due to macroeconomic imbalances of low economic growth, rising government debt and a volatile exchange rate (Thenuwara, 2019). Exchange rate volatility signifies the strain on Sri Lanka's international reserves. Foreign direct investment flows to Sri Lanka seem constrained. Compared to the Southeast Asian economies that continue to draw substantial investments, the South Asian bloc of economies, including Sri Lanka, are weighed down by their regional rigidities, high investment barriers, structural imbalances and necessary reforms that are either slow or reverse (Weerakoon, 2020), thus acting as disincentives for foreign investments.

It is important to examine the nexus between reserves, foreign exchange rate and inflation, given the exchange rate policy's role in the central bank's monetary policy framework and its stability objective. Reserves accumulation enables stabilizing the exchange rate in times of exchange rate stress. The Central Bank of Sri Lanka (CBSL) has had episodic depletion of its foreign reserves in defence of the rupee. In 2011–2012 and 2015–2016, interventions to stabilize the rupee cost Sri Lanka US \$4 billion and US \$2 billion, respectively, while

the intervention was needed to arrest the rupee's sharp depreciation during the latter part of 2018 and 2019 (Central Bank of Sri Lanka [CBSL], 2018, 2019a). In this context, if accumulated reserves play a significant role in inflationary pressure creation, the central bank's defence of the exchange rate may prove to be counterproductive.

The exchange rate regime and Irving Fischer's quantity theory of money (QTM) play an important role in CBSL monetary policy formation. CBSL carried out the monetary policy within a fixed-exchange-rate monetary policy framework during 1950–1977. It shifted to a managed-floating-exchange regime that supported a monetary targeting framework during 1982–2015. The monetary targeting framework and its current iteration, the enhanced monetary policy framework, use broad money supply—a monetary aggregate—as the nominal anchor, relying on the principles of QTM (Weerasinghe, 2017). In the present iteration of the monetary framework, CBSL uses an interest rate (the weighted average call money rate) as its operational target, perhaps as a preparation to shift to the flexible-inflation-targeting (FIT) framework (CBSL, 2019b; Weerasinghe, 2017).

Similarly, the stock of foreign reserves is equated to the stock of broad money, which is assumed to influence inflation. Although the New Keynesian Phillips curve (NKPC) and micro-founded dynamic stochastic general equilibrium (DSGE) models have at their roots the interest rate-based rule, this study is justified in using the QTM to gauge the monetary influence on inflation, as CBSL has linked its monetary policy formula to monetary aggregate albeit with sights set on shifting to FIT. Therefore, the extended use of QTM to determine the inflation–reserves nexus is deemed as a justifiable premise.

This study is a first for Sri Lanka, which attempts to bridge an important gap in identifying foreign reserves accumulation as a policy-driven factor that affects inflation in Sri Lanka. Previous empirical studies have focused on identifying the cost push–demand pull determinants of inflation in Sri Lanka (Deyshappriya, 2014; Kulatunge, 2012; Ratnasiri, 2009). However, surprisingly, these studies have not considered the influence of accumulated foreign reserves on inflation. Based on our findings and because reserves accumulation is a policy-driven objective, this article identifies foreign reserves as an important policy variable that should be included in future studies on inflation. This article has twin objectives: first, to observe the inflation–reserves nexus and identify the interrelationships between inflation, reserves and exchange rate and, second, to examine the magnitude and direction of the short-term and long-term impact of reserves build-up on inflation.

There are several noteworthy caveats to consider. Recent empirical evidence from a time-varying analysis shows some decoupling among the growth of broad money and other macroeconomic variables, such as GDP growth and inflation in Sri Lanka (Jegajeevan et al., 2019). One reason for CBSL to reconsider its monetary policy framework is also due to this variance. Such decoupling and variance were witnessed among sufficiently liquid financial markets during the height of the global financial crisis (Financial Crisis Inquiry Commission, USA, 2011; Han, 2016). The present article and its analysis should be observed in the context of weakening of the relationship between the variables.

This study's outcome is expected to provide a perspective towards the short-run and long-run impacts of foreign reserves accumulation and contribute to policy discussions relating to the core price stability mandate of the CBSL. Further, the findings would enrich discussions on inflation expectations, if the proposed shift to an FIT framework occurs in the future.

In attempting this, the current article is organized as follows. Following the introduction, the second section delves into the literature review, and the third section provides data and some preliminary observations on the relationship. The fourth section provides the economic rationale and methodology. The fifth section covers data analysis and empirical findings. The final section concludes with the results and their relevance to policy discussions.

Inflation and Foreign Reserves: Brief Review of the Literature

Reserves accumulation's impact on financial stability is well documented. However, not many discuss the inflationary influence of foreign reserves—the price stability impact. The foray into studying reserves accumulation and its impact on inflation commenced with Heller (1966, 1976), who looked into the global inflationary effect from worldwide reserves accumulation and estimated the optimum level of international reserves through comparing marginal benefits (Kashif et al., 2016). The QTM approach showed that the money supply's growth would cause the aggregate price levels to increase and contributed to the analysis of the nexus between international reserves and worldwide inflation (Heller, 1976; Laffer & Meiselman, (1975). The theory propounds that a rise in reserves, if not thoroughly sterilized, would increase the monetary base, which, coupled with the money multiplier, would cause an expansion of the total quantity of money, prompting inflation; this was empirically observed by Steiner (2017), who identified that Heller's worldwide inflation approach would not apply in the context of floating exchange rates. Steiner (2009, 2010, 2017) looked into the theoretical underpinnings and empirical observations relating to the inflation–reserves nexus, among stability concerns. Steiner's influence is critical in this area, and this study is one such beneficiary.

Further, several country studies in this area saw a geographic concentration of interest on reserves accumulation's effects. Foreign reserves accumulation continent-wise shows that by the end of 2011, Asian economies were at the forefront of central banks–led reserve build-up, with China and Japan leading in terms of the magnitude of reserves accumulated. Studies on several Asian and East Asian economies (Lin & Wang, 2005) and Vietnam (Phung Nguyen et al., 2019) both attribute to Steiner's model the structure by Heller engaging QTM. The African economies had, in general, taken steps to accumulate funds as a defensive strategy to act as an adequate safeguard during global economic downturns where international crisis management is either weak or absent (Akpan, 2016; Drama, 2016). Empirical studies have also been carried out for Pakistan (Chaudhry et al., 2011),

using autoregressive distributed lag (ARDL) measures, Tunisia (Bellamy, 2014) and Nigeria (Akpan, 2016). The European Union is covered in Chițu et al. (2019), and studies have also been carried out for China (Kashif et al., 2016; Narayan & Smyth, 2006).

The central bank's independence has often played a significant role in policy-making and management of foreign reserves build-up. A central bank's degree of autonomy in its objectives, as identified by Kydland and Prescott (1977), plays a deciding factor on the policy-led objectives such as accumulation and management of foreign reserves. External factors, such as the global financial crisis, also prompted central banks to build up reserves (Balls et al., 2016). As a panacea to outlast the crisis, the International Monetary Fund (IMF) also called for reserves build-up and buffer creation. In contrast, reserves adequacy was touted as a metric of a country's potential foreign-reserves liquidity that would act as a cushion to weather shocks (IMF, 2016). The global financial crisis drew attention to the central banks' role in stabilizing domestic and international financial systems and highlighted foreign reserves' role in such objectives (Cunningham & Friedrich, 2016; Han, 2016; IMF, 2019). It is arguable whether the foreign-reserve buffers pose as the sword of Damocles to the central banks tasked with the traditional stabilization goals. The emerging consensus was that much should be done by central banks to manage inflationary expectations when engaging the New Keynesian model based inflation targeting frameworks (Eichengreen et al., 2011; Vredin, 2015). Several related issues, such as global-financial-system stability (Balls et al., 2016), beggar-thy-neighbour policy spillovers that impact monetary policies of peers and cross-border capital flows, also played a predominant role in reserves build-up (Eichengreen et al., 2011; Leeper & Nason, 2014; Magud et al., 2018; Ostry et al., 2012). Increased capital inflows encourage the reserves buffer to build up in a country (Steiner, 2017). However, understanding the relationship between foreign reserves and the inflationary pressures phenomenon is vital to better manage such buffers in future.

Inflation and its determinants have often been subject to many literary discussions. Literature in this area focuses on modelling approaches offered by the long-run expectations-augmented Phillips curve (EAPC), the NKPC, with focus on the output gap and cost-push effects on inflation, the hybrid approach of backward- and forward-looking Phillips curve (Mankiw, 2001) and the monetary approach to inflation. While the latter is a backward-looking approach, EAPC and NKPC both favour the forward-looking expectations involved in the calculations. Many empirical studies have been based on these models, obtaining mixed results. According to Gali and Gertler (1999), NKPC provides an excellent first-approximation source of inflation in forward-looking theories. However, they did not rule out the usefulness of other methods and the efficacies in having a backward-looking methodology among the methods in determining inflation's causation. Influential work was done by Pesaran et al. (2001) on the ARDL technique, and QTM is the method that is engaged in the present work in assessing the nexus between inflation and the foreign reserves maintained by CBSL. Compared with NKPC, EAPC or real business cycle-based DSGE models, QTM is relied on in this article, as it is more suited to the Sri Lankan context. CBSL, relied in the past on a quantity theory based monetary targeting

framework, and since 2015 till recently on an enhanced monetary policy framework. Both use broad money stock as nominal indicative anchor. At present it conducts its monetary policy in line with a flexible inflation targeting (FIT) framework. Therefore, QTM based analysis in the present context is deemed more appropriate to observe the relationship between foreign reserves and inflation.

International Institutional Views on Reserves Accumulation, Management

Reserves function as a mirror reflecting the country's macroeconomic health to the outside world. Policy discussions by international organizations have contributed to the accumulation and efficient management of foreign reserves buffers. The Guideline of Foreign Exchange Reserve Management (IMF, 2001) identifies several objectives that could be achieved through maintaining and managing reserves, including: (a) supporting monetary and exchange rate management; (b) limiting external vulnerabilities through absorbing shocks during times of crisis; (c) providing market confidence in external obligations; (d) demonstrating domestic currency as being backed by external assets; (e) assisting the government in meeting its foreign exchange needs and external debt obligations; and (f) maintaining reserves for national disasters or emergencies. The Bank for International Settlement (BIS) in its Annual Economic Report (BIS, 2019) recognizes reserves accumulation as an integral part of the monetary framework, especially among the emerging market economies (EMEs) using inflation-targeting frameworks. Similarly, the European Central Bank adopts its foreign reserves policy accumulation and management based on the principles of liquidity, security and returns (Chițu et al., 2019).

Sri Lankan Literature on Inflation and Its Determinants, Reserves, Pass-through Exchange Rate

Several theoretical and empirical studies have shown inflation being influenced by both demand- and supply-side factors (Bandara, 1996; Colombage, 2005; Cooray, 2008; Deyshappriya, 2014; Jayawardena & Jayasinghe, 2016; Kulatunge, 2012; Nicholas, 1993; Ratnasiri, 2009; Rupananda, 1994; Weerasekara, 1992). Furthermore, the literature on inflation has discussed, *inter alia*, inflation persistence across exchange rate regimes (Harischandra, 2007), the magnitude of pass-through rates (Wimalasuriya, 2007), monetary policy impact on growth and inflation (Amarasekara, 2008), inflation–unemployment trade-off (Amarasekara & Bratsiotis, 2015), inflation–output trade-off (Kesavarajah & Middleditch, 2019) and the contribution of oil price (Perera, 2005) and other factors to the inflation in Sri Lanka (De Silva, 2017). Samaratunga and Perera (2015) looked at the demand and adequacy of international reserves in Sri Lanka. Wimalasuriya (2007) examined the passing through of exchange rates to Sri Lankan prices,

Table 1. Descriptive Statistics.

Variable	Mean	Median	Maximum	Minimum	Std. Dev.	Skewness	Kurtosis	Jarque-Bera	Prob.
CCPI	87.66311	90.10	134.6	40.10	28.31713	-0.235314	1.807705	14.10293	0.000866
CBR	4,865.187	5,611.75	8,573.17	1,288.28	2,075.055	-0.211928	1.602281	18.31058	0.000106
EXR	125.38	114.8	182.13	94.50	23.8149	0.839920	2.760386	24.7138	0.000004

Source: The authors.

namely the factor input prices, trade prices, wholesale producer prices and retail consumer prices.

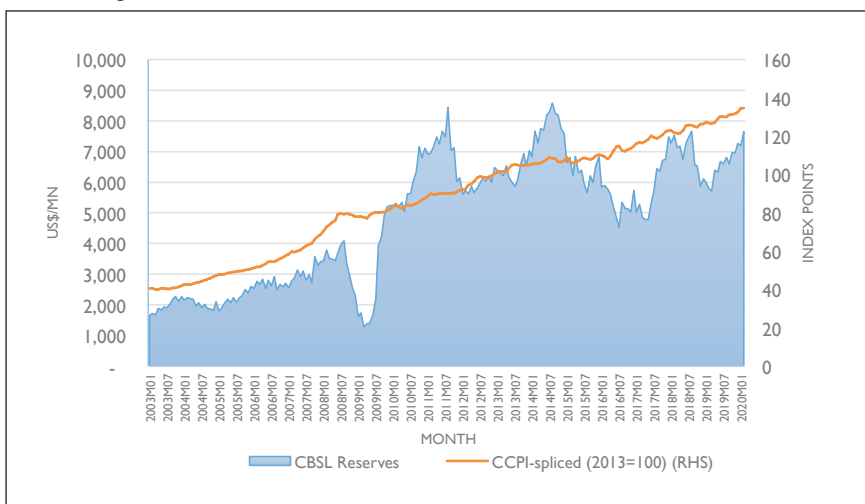


Figure 1. Central Bank Foreign Reserves and CCPI (2013 = 100).

Source: Central Bank of Sri Lanka (2018, 2019a).

Data and Preliminary Observations on the Relationships

The Colombo Consumer Price Index (CCPI), compiled since 1952, is the official index that measures movements in consumer prices in Sri Lanka. Headline inflation is measured in Sri Lanka using CCPI and the National Consumer Price Index (NCPI). This study relies on CCPI, due to its longer time series than NCPI, and uses monthly data from 2003M01 to 2020M02 (206 observations), with base (2013 = 100), spliced to capture base years 2002 and 2006–2007. Central bank reserves (CBR) are month-end foreign reserves maintained by CBSL in US dollar millions for the period from 2000M05 to 2020M02 (238 observations). Further, from 2000M01 to 2020M02, the average monthly LKR/USD exchange rate (EXR) is also used. Data were sourced from the CBSL. Table 1 shows descriptive statistics.

At the height of Sri Lanka's war against terrorism, CBR was the lowest at US\$1.29 billion in March 2009. The skewness and kurtosis figures show all variables as platykurtic (<3) and distributed normally. However, a close observation of the Jarque–Bera and probability values shows that the null is rejected, indicating a slight deviation from the normal distribution. All variables are used in their logarithmic forms. Figure 1 shows the time series trends of CCPI and CBR. The sudden drop in reserves between 2008 August and 2009 July is largely due to the conflict's intensity.

Methodology

Quantity Theory of Money and Net Foreign Assets

Irving Fisher's QTM states: $M \times V = P \times Y$

(M = money supply; V = velocity of money; P = average price level or the GDP deflator; and Y = aggregate output).

When depicted as rates of changes (denominated by the $\hat{}$ sign): $\hat{M} + \hat{V} = \pi + \hat{Y}$
Also, $\pi = \hat{M} + \hat{V} - \hat{Y}$. (1)

As money supply (M^s) is determined based on the money multiplier (m) and the money base, which consist of net domestic assets (NDA) and net foreign assets (NFA):

$$M^s = m \times (\text{NDA} + \text{NFA}). \quad (2)$$

Therefore, the rate of change in M can be expressed as:

$$\hat{M} = \hat{m} + \frac{\text{NDA}}{B} \cdot \widehat{\text{NDA}} + \frac{\text{NFA}}{B} \cdot \widehat{\text{NFA}}. \quad (3)$$

where B is the monetary base.

Combining Equation (3) with Equation (1):

$$\pi = \hat{m} + \frac{\text{NDA}}{B} \cdot \widehat{\text{NDA}} + \frac{\text{NFA}}{B} \cdot \widehat{\text{NFA}} + \hat{V} - \hat{Y}. \quad (4)$$

For a given economy, the velocity of money is constant; therefore, $\hat{V} = 0$.

$$\text{Then, } \pi = \hat{m} + \frac{\text{NDA}}{B} \cdot \widehat{\text{NDA}} + \frac{\text{NFA}}{B} \cdot \widehat{\text{NFA}} - \hat{Y}. \quad (5)$$

Steiner (2017), Equations (4) and (5) show inflation linked with changes to foreign reserves and domestic assets. Changes to NDA and NFA have inflationary consequences, all other things remaining equal.

It can be hypothesized that when the central bank fully sterilizes its domestic assets using open market operations (OMO), then at the extremity where

$$\frac{\text{NDA}}{B} \cdot \widehat{\text{NDA}} = -\frac{\text{NFA}}{B} \cdot \widehat{\text{NFA}}, \text{ inflation is unaffected.}$$

Based on the assumption that the central bank does not do NDA sterilization, then as $\Delta \text{NDA} = 0$, inflation (π) reflects changes to NFA/foreign reserves. Ergo, the change in foreign reserves (assets) would have a positive effect on inflation. Based on the above rationale, this study hypothesizes that the CBSL's foreign reserves indirectly impact the headline inflation reflected by CCPI in the short and the long run.

Therefore, this study's premises are:

H'₀: There is no long-run relationship between CBR, EXR and CCPI, *ceteris paribus*.

H'₁: There is a significant long-run relationship between CBR, EXR and CCPI, *ceteris paribus*.

Autoregressive Distributed Lag Model

This study engages an ARDL model, which uses standard least-squares regressions that include lags of both the dependent variable and explanatory variables as regressors (Greene, 2008). In an ARDL model, the dependent variable is depicted as a function of its lagged values, along with the current and lagged values of other exogenous variables of the model. In recent years, ARDL models have gained popularity as a method of examining cointegrating relationships between variables, through the work of Pesaran and Shin (1996) and Pesaran et al. (2001).

Several noteworthy preconditions should be adhered to when using an ARDL model. First, the series under consideration must be stationary, either in the level form, $I(0)$, or in the first-differenced form, $I(1)$. Second, as the model is sensitive to the number of lagged terms used, care should be exercised in selecting the lagged periods. Third, the error terms in the causality tests should remain unrelated. Fourth, for determining the long-run relationship, evidence of cointegration must be present. The diagnostic tests such as cumulative sum (CUSUM) tests, CUSUM-squared tests, normality tests and Lagrange multiplier (LM) tests are carried out to check the model validity. Variance decomposition tests are engaged to enrich the coefficient diagnosis, as appropriate, to observe the model fit and the impact of the variables on each other.

Often, the ARDL modelling approach is used in the dynamic econometric models such as autoregressive models that portray the time path of dependent variables concerning their past values:

$$Y_t = \alpha + \beta X_t + \gamma Y_{t-1} + u_t$$

In the Pesaran and Shin (1999) and Pesaran et al (2001) model:

$$\Delta Y_t = \beta_0 + \sum_{i=1}^n \beta_i \Delta y_{t-i} + \sum_{i=0}^n \delta_i \Delta x_{t-i} + \phi_1 y_{t-1} + \phi_2 x_{t-1} + \mu_t$$

Short-run effect

Long-run effect

disturbance

Accordingly, β_i and δ_i are the short-run coefficients, while ϕ_1 and ϕ_2 are the ARDL long-run coefficients. μ_t is the white noise. The ARDL is engaged here under three scenarios: (a) to determine the effect of CBR on inflation, (b) to determine the effect of EXR on inflation, and (c) to determine the combined effect of CBR and EXR on inflation.

First, CCPI behavior as a dependent variable against the exogenous variable of CBR at time t is used in the following form:

$$CCPI_t^d = \beta_0 + \beta_1 CBR_t + \varepsilon_t \quad (6)$$

β_0 is the constant, and ε is the stochastic disturbance term. The variables are depicted in their log forms as:

$$\ln(CCPI_t^d) = \beta_0 + \beta_1 \ln(CBR_t) + \varepsilon_{jt} \quad (7)$$

From Pesaran and Shin (1996) and Pesaran et. al. (2001):

$$\Delta Y_t = \beta_0 + \sum_{i=1}^n \beta_i \Delta y_{t-i} + \sum_{i=0}^{p,q} \delta_i \Delta x_{t-i} + \phi_1 y_{t-1} + \phi_2 x_{t-1} + \mu_t \quad (8)$$

$$\Delta \ln(\text{CCPI})_t = \beta_0 + \sum_{i=1}^n \beta_1 \Delta \ln(\text{CCPI})_{t-i} + \sum_{i=0}^n \delta_1 \Delta \ln(\text{CBR})_{t-i} +$$

$$\phi_1 \ln(\text{CCPI})_{t-1} + \phi_2 \ln(\text{CBR})_{t-1} + \mu_t. \quad (9)$$

(Null Hypothesis (H'_0): $H_0 = \phi_1 = \phi_2 = 0$)

The ARDL model provides a substitute test for examining the long-run relationship regardless of whether the underlying variables are I(0), I(1) or fractionally integrated. The resulting F -statistic of the ARDL bounds-testing approach provides critical values for both the lower bound, I(0), and the upper bound, I(1). A higher-than-the-upper-bound F -statistic would indicate the presence of a long-run effect, which then is to be further verified using the error correction model (ECM) version of the modified ARDL (as employed in Chaudhry et al. [2011]) and ordinary least squares (OLS) for the existence of the short-run relationship. While using the ARDL approach, the standard correlation methods, Johansen cointegration framework, unit root testing, etc., are also addressed.

Data Analysis and Empirical Findings

Estimation of Unit Root and Robustness

Unit-root estimation is important in the ARDL technique to ascertain variables that remain integrated within the level form and/or I(I). Following Pesaran et. al. (2001), Dickey–Fuller generalized least squares (DF–GLS) test with Akaike information criterion (AIC), Phillips–Perron (PP) and Kwiatkowski–Phillips–Schmidt–Shin (KPSS) tests are used on a relatively large sample size for Sri Lanka. The augmented Dickey–Fuller test used by Chaudhry et al. (2011) is less relied upon (De Jong et al., 1992), (Harris et al., 2003). The results show all variables integrated of order 1 (Table 2), prompting us to engage the ARDL bounds-testing approach to examine the long-run relationships of CCPI with CBR and EXR individually and that of CCPI with CBR and EXR together.

ARDL Model Using the Bounds Test Approach

The ARDL bounds-testing approach identifies the presence of long-run cointegration between LCCPI and LCBR and between LCCPI and LEX (hereafter, LEX represents log value of EXR) (Table 3).

Table 2. Stationarity of Variables (Intercept, Trend and Intercept).

Variable	Intercept/ Trend and Intercept	DF-GLS (AIC)		PP		KPSS		Order of Integration
		Level	1st Difference	Level	1st Difference	Level	1st Difference	
LCCPI	Trend, intercept	-0.671001	-3.146237**	-0.699598	-9.818861***	0.423147	0.109506***	I(1)
LCBR	Trend, intercept	-3.666097***	-6.938416***	-2.699447	-17.09591***	0.294341	0.032001***	I(1)
LEX	Trend, intercept	-2.846896	-14.14620***	-3.053317	-14.27638***	0.176688*	0.104512***	I(1)

Source: The authors.

Note: *, ** and *** Indicate significance at the 1%, 5% and 10% critical value levels, respectively.

Table 3. The Results of Cointegration Test

Panel I:ARDL bounds testing to cointegration (restricted constant without trend)					Asymptotic Critical Value Bounds for F-Statistic		
ARDL Model	Dependent Variable	Independent Variable	Computed F-Statistic		5% Level		1% Level
					I(0)	I(1)	I(0)
ARDL(3,0)	ΔLCCPI	LCBR	13.7268***		3.62	4.16	4.94
ARDL(3,0)	ΔLCCPI	LEX	13.9744***		3.62	4.16	4.94
ARDL(3,0,0)	ΔLCCPI	LCBR,LEX	10.47032***		3.10	3.87	4.13

Panel II: Diagnostic tests			
Model (dep. var./ ind. var.)	LCCPI/LCBR	LCCPI/LEX	LCCPI/LCBR, LEX
R^2	0.999463	0.999465	0.999465
Adjusted- R^2	0.999452	0.999454	0.999452
F -statistic	92152.47***	92438.83***	73629.94***
Breusch–Godfrey LM test	0.761711 (0.4682)	0.688215 (0.5037)	0.733545 (0.4815)
ARCH LM	0.304583 (0.8221)	0.333331 (0.8013)	0.372360 (0.5424)
White heteroscedasticity	0.961864 (0.4827)	0.798659 (0.6413)	0.747383 (0.7508)
Ramsey RESET	1.546637 (0.1236)	3.988416 (0.0001)	4.657723 (0.0000)
CUSUM	Stable	Stable	Stable
CUSUM squared	Stable	Stable	Stable

Source: Authors' calculation based on the ARDL models.

Note: *** denote statistical significance at the critical value level of 1%. Figures in parentheses represent the probability values of the diagnostic tests.

As shown in panel 1 of Table 3, F -statistic greater than the critical values show ARDL bounds-testing significant at the 1% level, which implies that there is cointegration (or long-term relationship) of LCCPI with both LEX and LCBR. Panel II of Table 3 shows the diagnostic test carried out to check whether the results are free from any serial correlation, White heteroscedasticity, misspecification and non-linearity of residuals. The CUSUM and CUSUM-squared tests find parameter testing consistent within the critical bounds of 5%. Assured by these outcomes, we then proceed to analyse the short-run dynamics using the ECM version of the ARDL model.

Error Correction Model Results in the Short-run Effect

The ECM shows the error correction term (ECT) in the CBR's and EXR's impacts on the CCPI as negative and significant (Table 4). ECT indicates speed of adjustment and negative sign indicates a convergence from short run to long run. The Coint.Eq. in Table 4 depicts the speed of adjustment to equilibrium in every period in the presence of the cointegrating relationship between CCPI, CBR and EXR (shown in its log forms LCCPI, LCBR and LEX). To be more precise, in an event of disequilibrium during the long run, inflation as referenced by CCPI will adjust to equilibrium as short run adjustments at a speed of 0.5% and 0.7%, in the presence of CBR and EXR, respectively. But when reserves combined with exchange rate, speed of adjustment of CCPI is marginally quicker at 0.9%.

The ECT defines how quickly each variable's deviation in the long-run equation is corrected gradually towards the equilibrium level through a series of partial short-run adjustments (Chaudhry et al., 2011; Kulatunge, 2012). Considering the inflation–reserves nexus, the ECT for CBR depicts that 0.5% of the long-run

adjustment of CCPI is envisaged from the partial short-run adjustment of CBR every month. The magnitude of the ECT is statistically significant. The negative sign implies a convergence from the short run to the longer term and that the term is stable. This means the speed of correction in the long-run/short-run adjustment is 0.5% per month if the economy suffers unexpected inflation due to uneven variations in the central bank foreign reserves. In other words, the 0.5% disequilibrium in inflation from a shock in the previous month due to the central bank's foreign reserves converges back to the long-run equilibrium in the current month. Although this adjustment is slow, we contend that the impact cannot be disregarded as 'insignificant'.

We attempt to find further empirical confirmation of CCPI's reaction to both EXR and CBR in the short run. As there is a cointegrating relation between the variables based on the outcome of the ARDL cointegration tests, the null hypothesis, (H_0 : $H_0 = \phi_1 = \phi_2 = 0$), can be rejected. However, the short-run relation between LCCPI and LCBR and LEX is depicted through modifying Equation (9) to accommodate ECT:

$$\Delta \ln(\text{CCPI})_t = \phi_0 + \sum_{i=1}^n \alpha_1 \Delta \ln(\text{CCPI})_{t-i} + \sum_{i=0}^n \phi_1 \Delta \ln(\text{CBR})_{t-i} + \phi_2 \Delta \ln(\text{EX})_{t-i} + \gamma \text{ECT}_{t-1} + \mu_t \quad (9b)$$

(Alternate hypothesis [H_1]: $H_1 \neq \alpha_1 \neq \phi_1 \neq \phi_2 \neq 0 \neq Y$).

Based on Equation (9b), the cointegration equation is described as:

$$\text{ECT}_{t-1} = \Delta \ln(\text{CCPI})_{t-1} - \phi_0 - \phi_1 \ln(\text{CBR})_{t-1} - \phi_2 \ln(\text{EX})_{t-1}$$

$$\text{ECT}_{t-1} = 1.000 \text{ LCCPI}_{t-1} - 0.5137 - 0.1024 \text{ LCBR}_{t-1} - 0.7389 \text{ LEX}_{t-1}$$

This explains the last-period deviation from the long-run equilibrium of 0.1% of the central bank accumulated reserves, and 0.73% of the exchange rate explains the short-run dynamics of LCCPI. However, neither coefficient is statistically significant. The short-run elasticity obtained from the ARDL is also not statistically significant. For the combined effect of EXR and CBR on CCPI, the ARDL short-run elasticities are statistically insignificant.

Table 4. Short-run Results.

Variable	LCBR	LEX	LCBR and LEX
$\Delta \text{LCCPI}(-1)$	0.373393***	0.372693***	0.372485***
$\Delta \text{LCCPI}(-2)$	-0.137030**	-0.133664*	-0.133992*
Coint. Eq.(ECT)	-0.005094***	-0.007318***	-0.008826***

Source: Authors' calculation based on the ARDL models.

Note: ***, ** and * Denote statistical significance at the critical value levels 1%, 5% and 10%, respectively.

Vector Autoregression Model

To test the robustness of our results from the ARDL model, we engage a vector autoregression (VAR) approach using LCCPI, LCBR and LEX. Three lags are selected for VAR using the lag selection criteria. The VAR results are subjected to the maximum-likelihood procedure by Johansen (1988) and Johansen and Juselius (1990) to determine the cointegration between LCCPI and LCBR. The trace test's null hypothesis assumes that the number of distinct cointegrating vectors is less than or equal to (q) against a general unrestricted alternative ($q = r$). The trace statistic tests the null hypothesis of r cointegrating relations against the alternative of k cointegrating relations, where k is the number of endogenous variables, for $r = 0, 1, \dots, k - 1$. The alternative k of cointegrating relations corresponds to the case where none of the series has a unit root, and a stationary VAR may be specified in terms of the levels of all of the series. This statistic is specified as follows:

$$\lambda_{\text{trace}}(r) = -T \sum_{i=r+1}^p \ln(1 - \lambda_i)$$

($\lambda_{r+1} \dots \lambda_n$ are the smallest-value eigenvectors [$p - r$]).

Similarly, the maximum-eigenvalue test (λ_{max}) is calculated as follows:

$$\lambda_{\text{max}}(r, r+1) = -T \ln(1 + \lambda_{r+1})$$

This concerns the null hypothesis of r cointegrating relations against the alternative of $(r + 1)$ cointegrating relations. The results from the statistical tests, the trace test (λ_{trace}) and the maximum-eigenvalue test (λ_{max}) from a VAR-based Johansen cointegration, are presented in Table 5.

Table 5. Johansen Cointegration Test.

Panel I: VAR-based unrestricted cointegration rank test (LCCPI and LCBR)						
Trace	Maximum Eigenvalue					
Hypothesized						
No. of			5%			5%
Cointegrated	Trace	Critical		Trace	Critical	
Equations	Statistic	Value	Prob.**	Statistic	Value	Prob.***
None ($r = 0$)	15.70971*	15.49471	0.0464	9.276296	14.26460	0.2639
At most 1 ($r \leq 1$)	6.433418*	3.841466	0.0112	6.433418*	3.841466	0.0112
Panel II: VAR-based unrestricted cointegration rank test with the trend (LCCPI, LCBR and LEX)						
None ($r = 0$)	36.41622*	29.79707	0.0075	21.92671*	21.13162	0.0386
At most 1 ($r \leq 1$)	14.48951	15.49471	0.0704	13.94321	14.26460	0.0561
At most 2 ($r \leq 2$)	0.546297	3.841465	0.4598	0.546297	3.841465	0.4598

Source: The authors.

Notes: *Denotes rejection of the hypothesis at the 0.05 level.

**The trace test indicates two cointegrating equations at the 0.05 level.

***The maximum-eigenvalue test indicates cointegration at the 0.05 level.

The VAR-based cointegration tests' results (Table 5) show the presence of cointegration between LCCPI and LCBR, and between LCCPI, LCBR and LEX. These results confirm the presence of cointegration, suggesting long-run relationships between the discussed variables.

Causality Test

We carry out Wald and Granger causality tests to determine the long-run/short-run causality between the variables (see Appendix A for test results). The Wald test is engaged for the null hypothesis, where the ECT's cointegration is equated to 0 and used as a null representation. Both tests signal some feedback from CCPI to CBR, while the Granger causality test further suggests a two-way feedback between CBR and EXR. Therefore, the causality tests are treated as indeterministic in estimating the causal direction between CBR and CCPI and EXR.

Variance Decomposition

This study engages variance decomposition (VD) analysis to ascertain the relative importance of random shock (or innovation) affecting each variable in the VAR. In other words, VD helps quantify the proportion of variation of the dependent variable explained by each of the independent variables. Table 6 explains the variations among the variables due to single-standard deviation innovation. It seems that a significant share of variation in CCPI is explained by itself, where the 1st-month and 12th-month variations in CCPI attributed by CCPI are at 100.00% and 99.08%, respectively. LCBR and LEX explain less than 1% variance. However, the impacts of CBR and LEX on CCPI increase in the long run, with CBR accounting for more than half a percentage point of CCPI from the 5th month onwards. The results suggest some impact emanating from EXR being more than 10% on EXR from CBR from the 6th month onwards. Overall, it seems that CBR has a rather lukewarm impact on CCPI, compared to the significant share of variation in consumer prices explained by CCPI itself. However, these results reinforce the existence of a nexus between the variables and the long-run cointegration.

Table 6. Variance Decomposition Analysis.

Period (months)	VD of LCCPI			VD of LCBR			VD of LEX		
	LCCPI	LCBR	LEX	LCCPI	LCBR	LEX	LCCPI	LCBR	LEX
1	100.0	0.00	0.00	0.042	99.958	0.00	0.866	0.813	98.321
2	99.65	0.29	0.05	0.255	98.89	0.855	0.355	3.866	95.778
3	99.59	0.31	0.10	0.798	98.69	0.512	0.214	6.046	93.74
4	99.47	0.43	0.096	1.719	97.87	0.407	0.203	8.264	91.53
5	99.42	0.48	0.098	2.479	96.89	0.628	0.238	9.726	90.04
6	99.34	0.56	0.101	3.316	95.91	0.773	0.284	10.770	88.95
7	99.29	0.59	0.116	4.023	94.94	1.034	0.331	11.293	88.38
8	99.23	0.63	0.135	4.748	94.03	1.223	0.369	11.492	88.14
9	99.19	0.64	0.162	5.395	93.18	1.422	0.397	11.403	88.20
10	99.16	0.65	0.194	6.021	92.41	1.565	0.414	11.142	88.45
11	99.12	0.64	0.234	6.580	91.73	1.688	0.420	10.758	88.823
12	99.08	0.63	0.280	7.091	91.14	1.769	0.417	10.314	89.269

Source: The authors.

Outcome When the Model was Extended to Include the Foreign Exchange Rate

Giving heed to Kulatunge's (2012) and Ratnasiri's (2009) inclusion of the exchange rate as an inflation determinant, the article extends the model to include EXR. However, as seen in panel II of Table 3 and Table 5, the results were not conclusive when the robustness and diagnostic tests were carried out for the extended model. The extended model ARDL(3, 0, 0) used to test the effect on CCPI from CBR and EXR was not statistically significant. There were some indications of long-run cointegration between logarithmic values of EXR and CCPI. However, neither the VAR nor the vector error correction model (VECM) approach shows a meaningful short-run relationship with statistical significance between the three variables. This confirms the absence of a convincing short-run relationship (Appendices 1 and 2).

Furthermore, the VAR estimation is extended to impulse response functions to examine how a single-standard deviation shock applied to each variable impacts the others. The impulse responses for LCBR, LCCPI and LEX verify the earlier outcome. The response of CCPI to CBR is more or less similar to that of CCPI's response to LEX, albeit at a lower degree, as verified by the bounds and VD estimates. The results are provided in Appendix 3.

Conclusion and Policy Implications

This article's objective was to identify a nexus between the foreign reserves maintained by CBSL and the inflation as specified by the Colombo Consumer Price Index. Using the ARDL bounds-testing approach, we were able to determine the presence of a long-run relation between the two variables. The robustness of the ARDL outcome was checked using VAR, and several diagnostic tests were used to reinforce these results further. Initial causality tests showed two-way feedback between CBR and EXR, while the CCPI-to-CBR feedback suggests the need for further tests to study the relationship more conclusively.

Although the exchange rate is a determinant of monetary policy and inflation, this study did not find a significant impact of foreign exchange rate on inflation when coupled with foreign reserves. However, our VAR and VD analyses show some influence on foreign reserves. Furthermore, the results show some short-term effects on inflation and foreign reserves contributing to inflation beyond a 10-month period. These results in isolation may seem negligible, but their impact cannot be brushed away as insignificant without further research. The authors acknowledge that the bivariate model used here may not single out the reserves-inflation nexus in isolation from other multifaceted determinants of inflation identified through many previous studies. It is also acknowledged that the nexus between inflation and international reserves proves to be more complex in the presence of other determinants of inflation. This research gap should be addressed in future researches through analyses in the context of other determinants of inflation.

The results may prompt policymakers to favourably consider the accumulation of foreign currency reserves as a policy, while being aware of any future implications on inflation and inflationary expectations. This is important for any central bank that aims to create a buffer and expects to credibly forecast its inflation expectations primarily to cater to its stability mandates. More specifically, CBSL would benefit from this study, as it poses several questions to its reserves accumulation policies. Further, accurate forecasts of inflationary pressures would be essential to a monetary policy framework as that of Sri Lanka, as it aims to shift to an inflation-targeting framework in the near future. The results may prove to invoke new thinking into the interplay between foreign reserves and inflation and the former's interaction with the exchange rate.

It is believed that the results of this study would draw attention to the effects that could emanate from foreign reserves accumulation. They provide a new perspective on the agreement and functions of sovereign wealth funds. The current study has been done for a small, emerging open market economy. If it is further refined and extended to advanced economies, and coupled with other determinants of inflation, the findings may be of further interest to both monetary and fiscal policymakers, as well as to sovereign wealth fund practitioners. Economies should be cautious of possible outcomes as pointed out by this study which may foreshadow some of the benefits of foreign reserve buffers. This is especially important where policymakers try to assess any prolonged, unwanted inflationary impact, which may prove to be counterproductive to the central bank's price stability mandate, and where policies on capital control and macroprudential measures may also need to be resorted to at length.

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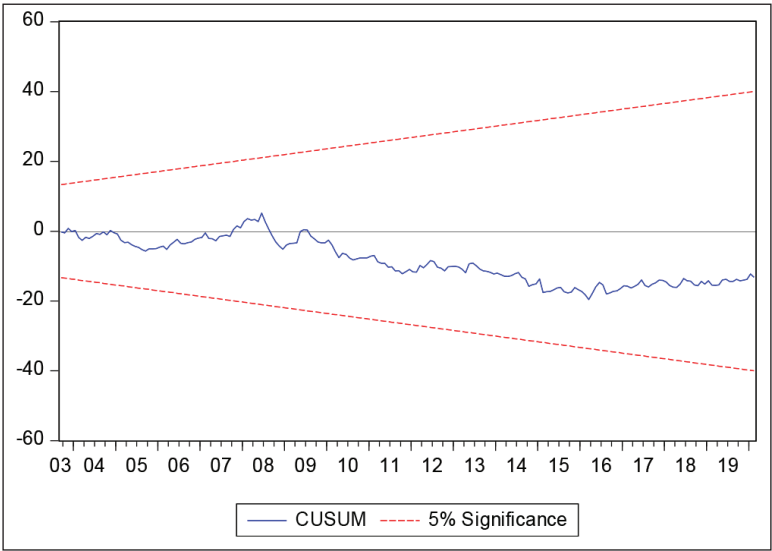
Ayesh Ariyasinghe  <https://orcid.org/0000-0003-4454-917X>

Appendix A

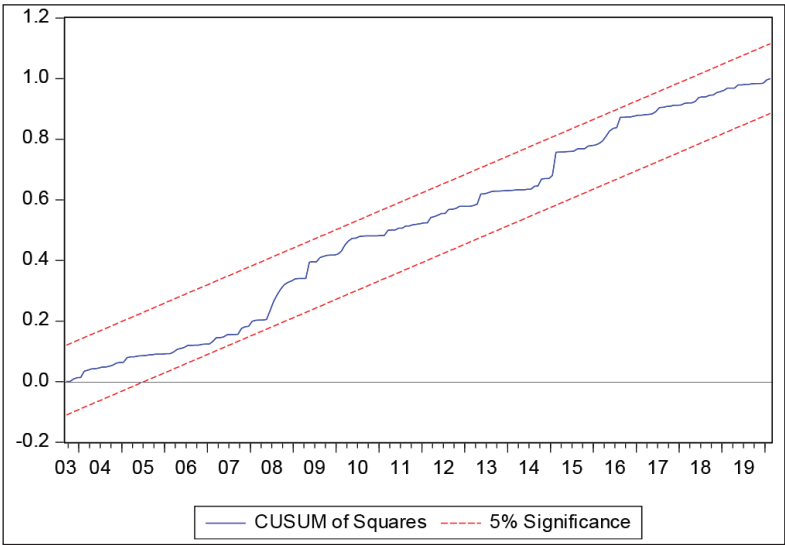
Diagnostic Tests

Dependent variable: CCPI
CUSUM test

Independent variable: CBR

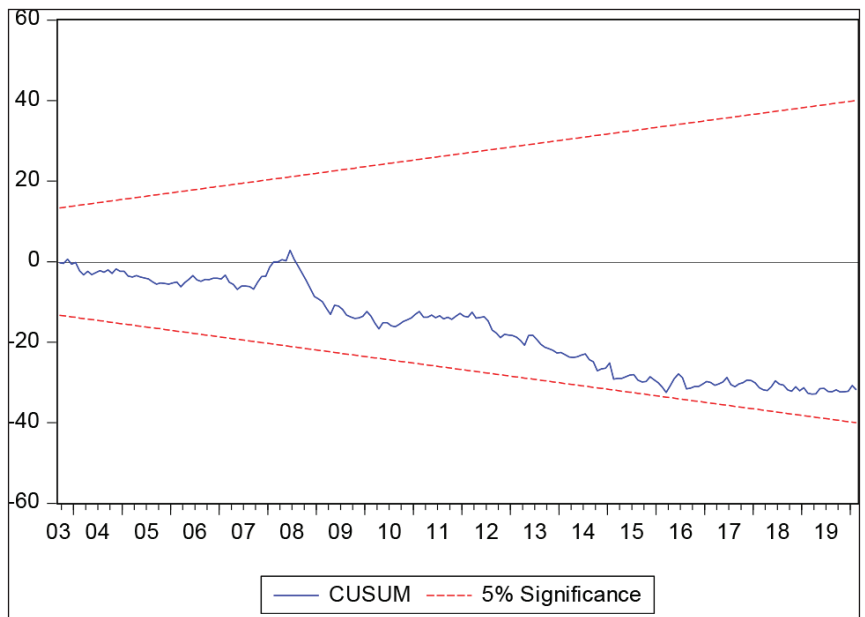


CUSUM-squared test

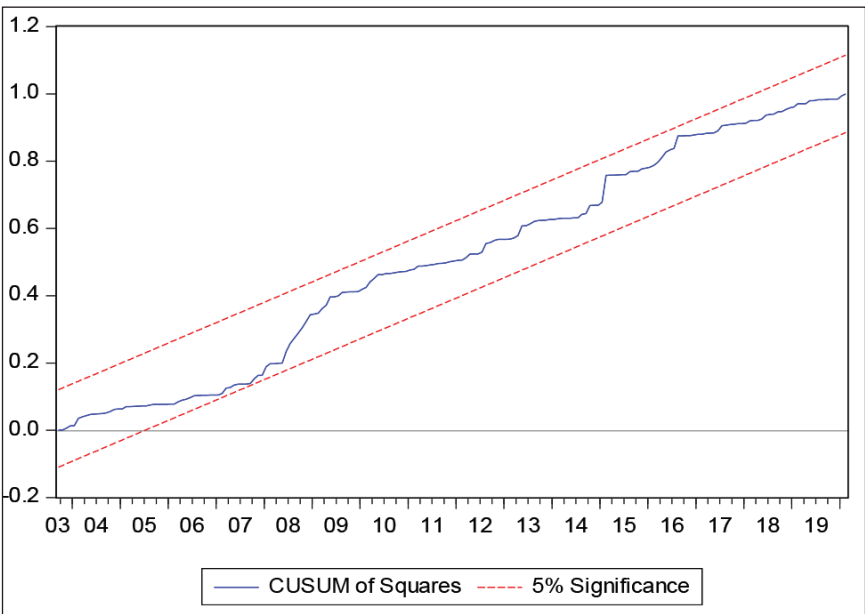


Dependent variable: CCPI
CUSUM test

Independent variable: EXR

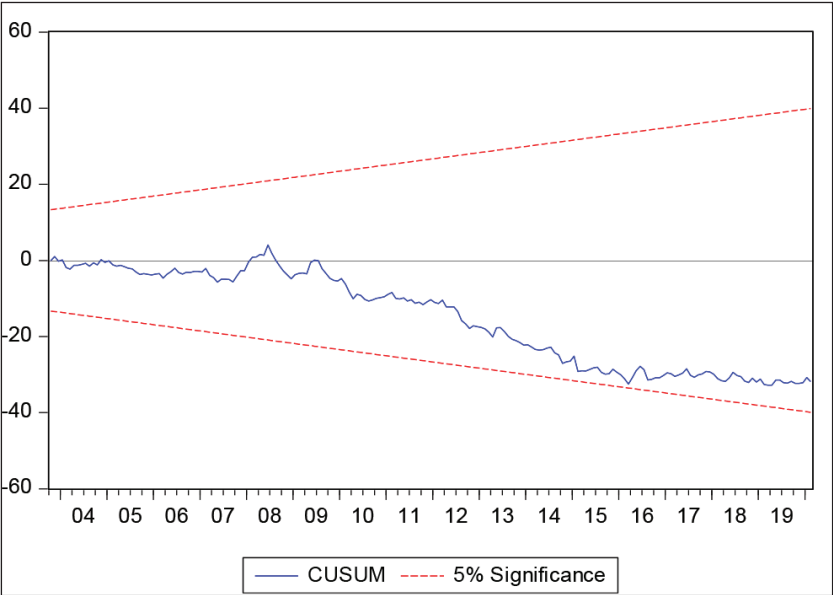


CUSUM-squared test

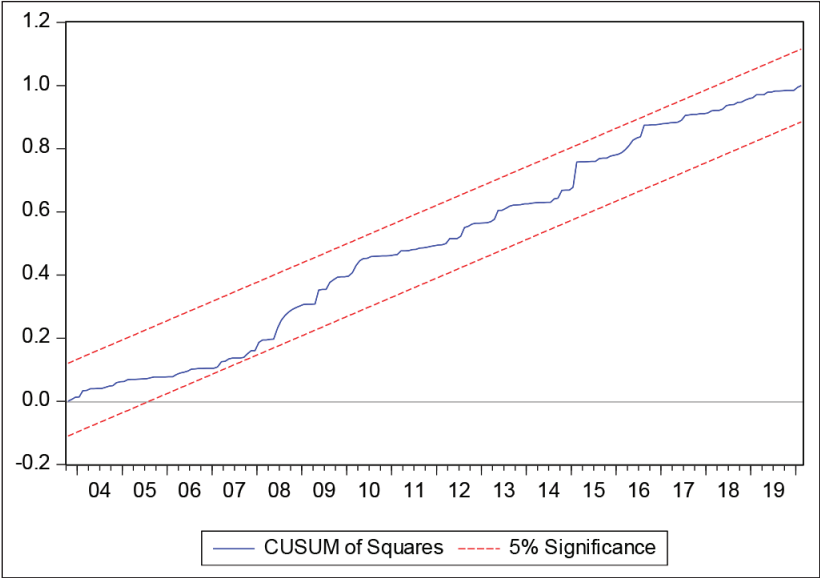


Dependent variable: CCPI
CUSUM test

Independent variables: CBR, EXR



CUSUM-squared test



Causality Tests

Granger Causality Test

VAR based on three optimum lags: LCCPI, LCBR and LEX

VAR Granger causality/block exogeneity Wald tests			
Date: 10/25/20; Time: 12:15			
Sample: 2000M01–2020M12			
Included observations: 203			
Dependent variable: LCCPI			
Excluded	Chi-sq.	df	Prob.
LCBR	3.003011	3	0.3912
LEX	0.826070	3	0.8432
All	3.847736	6	0.6973
Dependent variable: LCBR			
Excluded	Chi-sq.	df	Prob.
LCCPI	10.40423	3	0.0154
LEX	10.09035	3	0.0178
All	25.90213	6	0.0002
Dependent variable: LEX			
Excluded	Chi-sq.	df	Prob.
LCCPI	2.170045	3	0.5379
LCBR	11.63624	3	0.0087
All	15.13403	6	0.0192

Wald Test

Wald Test Outcomes

Null Hypothesis	F-stat.	Prob.	Outcome
CBR does not Granger-cause CCPI in the long run	1.427695	0.2337	Cannot reject null
CBR does not Granger-cause CCPI in the short run	0.625409	0.7558	Cannot reject null
CCPI does not Granger-cause CBR in the long run	6.766667	0.0101	Null is rejected
CCPI does not Granger-cause CBR in the short run	2.411939	0.0170	Null is rejected

Appendix I. Impact of Exchange Rate on CCPI.

Dependent variable: LCCPI

Method: ARDL

Date: 10/15/20 ;time: 15:22

Sample (adjusted): 2003M04–2020M02

Included observations: 203 after adjustments

Maximum dependent lags: 4 (automatic selection)

Model selection method: Akaike information criterion (AIC)

Dynamic regressors (4 lags, automatic): LEX

Fixed regressors: C

Number of models evaluated: 20

Selected model: ARDL(3, 0)

Note: The final equation sample is larger than the selection sample.

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LCCPI(−1)	1.365409	0.069959	19.51735	0.0000
LCCPI(−2)	−0.506393	0.114332	−4.429130	0.0000
LCCPI(−3)	0.133687	0.069529	1.922750	0.0559
LEX	0.005565	0.007113	0.782454	0.4349
C	0.009951	0.021777	0.456968	0.6482
R-squared	0.999465	Mean dependent var.		4.423064
Adjusted R-squared	0.999454	SD dependent var.		0.357650
S.E. of regression	0.008357	Akaike info criterion		−6.707013
Sum-squared resid.	0.013830	Schwarz criterion		−6.625407
Log likelihood	685.7618	Hannan–Quinn criterion		−6.673998
F-statistic	92433.65	Durbin–Watson stat.		2.009775
Prob. (F-statistic)	0.000000			

Note: *p-Values and any subsequent tests do not account for model selection.**ARDL Long-run Form and Bounds Test**

Dependent variable: D(LCCPI)

Selected model: ARDL(3, 0)

Case 2: restricted constant and no trend

Date: 10/15/20; time: 15:23

Sample: 2000M01–2020M03

Included observations: 203

Conditional error correction regression

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.009951	0.021777	0.456968	0.6482
LCCPI(-1)*	-0.007296	0.003557	-2.051174	0.0416
LEX**	0.005565	0.007113	0.782454	0.4349
D(LCCPI(-1))	0.372705	0.069601	5.354897	0.0000
D(LCCPI(-2))	-0.133687	0.069529	-1.922750	0.0559
**Variable interpreted as $Z = Z(-1) + D(Z)$.				
Levels equation				
Case 2: restricted constant and no trend				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
LEX	0.762753	0.670141	1.138197	0.2564
C	1.363874	3.462597	0.393888	0.6941
$EC = LCCPI - (0.7628 \times LEX + 1.3639)$				
F-bounds Test		Null Hypothesis: No Levels Relationship		
Test Statistic	Value	Significance Level	I(0)	I(1)
		Asymptotic: n = 1000		
F-statistic	13.96992	10%	3.02	3.51
k	1	5%	3.62	4.16
		2.5%	4.18	4.79
		1%	4.94	5.58
Actual sample size	203	Finite sample: n = 80		
		10%	3.113	3.61
		5%	3.74	4.303
		1%	5.157	5.917

ARDL Error Correction Regression

Dependent variable: D(LCCPI)

Selected model: ARDL(3, 0)

Case 2: restricted constant and no trend

Date: 10/15/20; time: 15:23

Sample: 2000M01–2020M03

Included observations: 203

ECM regression

Case 2: restricted constant and no trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LCCPI(-1))	0.372705	0.069176	5.387809	0.0000
D(LCCPI(-2))	-0.133687	0.068762	-1.944201	0.0533
CointEq(-1)*	-0.007296	0.001121	-6.506389	0.0000
R-squared	0.178508	Mean dependent var.		0.005965
Adjusted R-squared	0.170293	SD dependent var.		0.009129
S.E. of regression	0.008316	Akaike information criterion		-6.726717
Sum-squared resid.	0.013830	Schwarz criterion		-6.677754
Log likelihood	685.7618	Hannan–Quinn criterion		-6.706908
Durbin–Watson stat.	2.009775			

Note: *p-Value incompatible with t-bounds distribution.

F-bounds Test		Null Hypothesis: No Levels Relationship		
Test Statistic	Value	Significance Level	I(0)	I(1)
F-statistic	13.96992	10%	3.02	3.51
k	1	5%	3.62	4.16
		2.5%	4.18	4.79
		1%	4.94	5.58

Impact of Exchange Rate and CBR on CCPI

Dependent variable: LCCPI

Method: ARDL

Date: 10/15/20; time: 15:25

Sample (adjusted): 2003M04–2020M02

Included observations: 203 after adjustments

Maximum dependent lags: 4 (automatic selection)

Model selection method: Akaike information criterion (AIC)

Dynamic regressors (4 lags, automatic): LCBR LEX

Fixed regressors: C

Number of models evaluated: 100

Selected model: ARDL(3, 0, 0)

Note: The final equation sample is larger than the selection sample.

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LCCPI(-1)	1.363705	0.070261	19.40916	0.0000
LCCPI(-2)	-0.506513	0.114582	-4.420514	0.0000
LCCPI(-3)	0.134013	0.069687	1.923083	0.0559
LCBR	0.000901	0.002422	0.371927	0.7103

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
LEX	0.006499	0.007557	0.859952	0.3909
C	0.004518	0.026263	0.172027	0.8636
R-squared	0.999465	Mean dependent var.		4.423064
Adjusted R-squared	0.999452	SD dependent var.		0.357650
S.E. of regression	0.008376	Akaike information criterion		-6.697863
Sum-squared resid.	0.013820	Schwarz criterion		-6.599935
Log likelihood	685.8330	Hannan–Quinn criterion		-6.658245
F-statistic	73625.14	Durbin–Watson stat.		2.011088
Prob. (F-statistic)	0.000000			

Note: *p-Values and any subsequent tests do not account for model selection.

ARDL Long-run Form and Bounds Test

Dependent variable: D(LCCPI)

Selected model: ARDL(3, 0, 0)

Case 2: restricted constant and no trend

Date: 10/15/20; time: 15:25

Sample: 2000M01–2020M03

Included observations: 203

Conditional error correction regression

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.004518	0.026263	0.172027	0.8636
LCCPI(-1)*	-0.008795	0.005379	-1.634903	0.1037
LCBR**	0.000901	0.002422	0.371927	0.7103
LEX**	0.006499	0.007557	0.859952	0.3909
D(LCCPI(-1))	0.372500	0.069755	5.340121	0.0000
D(LCCPI(-2))	-0.134013	0.069687	-1.923083	0.0559

Notes: *p-Value incompatible with t-bounds distribution.

**Variable interpreted as $Z = Z(-1) + D(Z)$.

Levels equation

Case 2: restricted constant and no trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
LCBR	0.102444	0.232259	0.441078	0.6596
LEX	0.738933	0.566753	1.303801	0.1938
C	0.513709	3.238078	0.158646	0.8741
EC = LCCPI - (0.1024 × LCBR + 0.7389 × LEX + 0.5137)				

F-bounds Test		Null Hypothesis: No Levels Relationship		
Test Statistic	Value	Significance Level	I(0)	I(1)
Asymptotic: n = 1000				
F-statistic	10.46643	10%	2.63	3.35
k	2	5%	3.1	3.87
		2.5%	3.55	4.38
		1%	4.13	5
Actual sample size		Finite sample: n = 80		
	203	10	2.713	3.453
		5	3.235	4.053
		1	4.358	5.393

ARDL Error Correction Regression

Dependent variable: D(LCCPI)

Selected model: ARDL(3, 0, 0)

Case 2: restricted constant and no trend

Date: 10/15/20; time: 15:26

Sample: 2000M01–2020M03

Included observations: 203

ECM regression

Case 2: restricted constant and no trend

Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(LCCPI(-1))	0.372500	0.069147	5.387038	0.0000
D(LCCPI(-2))	-0.134013	0.068742	-1.949514	0.0527
CointEq(-1)*	-0.008795	0.001349	-6.519453	0.0000
R-squared	0.179085	Mean dependent var.		0.005965
Adjusted R-squared	0.170875	SD dependent var.		0.009129
S.E. of regression	0.008313	Akaike information criterion		-6.727419
Sum-squared resid.	0.013820	Schwarz criterion		-6.678456
Log likelihood	685.8330	Hannan–Quinn criterion		-6.707610
Durbin–Watson stat.	2.011088			

Note: *p-Value incompatible with t-bounds distribution.

F-bounds Test		Null Hypothesis: No Levels Relationship		
Test Statistic	Value	Significance Level	I(0)	I(1)
F-statistic	10.46643	10%	2.63	3.35
k	2	5%	3.1	3.87
		2.5%	3.55	4.38
		1%	4.13	5

Appendix 2. Vector Error Correction Model.

LCCPI and LCBR

Vector autoregression estimates

Date: 10/15/20; time: 15:37

Sample (adjusted): 2003M09–2020M02

Included observations: 198 after adjustments

Note: Standard errors in () and t-statistic in []

	LCCPI	LCBR
LCCPI(–1)	1.328841 (0.07377) [18.0139]	1.129598 (0.64496) [1.75141]
LCCPI(–2)	–0.461988 (0.12386) [–3.72986]	–0.384791 (1.08295) [–0.35532]
LCCPI(–3)	0.009495 (0.12872) [0.07376]	0.158037 (1.12546) [0.14042]
LCCPI(–4)	0.253534 (0.12851) [1.97294]	–1.247848 (1.12355) [–1.11063]
LCCPI(–5)	–0.087177 (0.12861) [–0.67782]	–0.510514 (1.12450) [–0.45399]
LCCPI(–6)	–0.002680 (0.12864) [–0.02083]	1.291480 (1.12472) [1.14827]
LCCPI(–7)	0.086792 (0.12386) [0.70074]	–2.470126 (1.08291) [–2.28101]
LCCPI(–8)	–0.131859	2.113021

	LCCPI	LCBR
	(0.07317)	(0.63973)
	[-1.80212]	[3.30299]
LCBR(-1)	0.010076	1.018557
	(0.00808)	(0.07062)
	[1.24743]	[14.4228]
LCBR(-2)	-0.013929	0.233672
	(0.01144)	(0.10006)
	[-1.21705]	[2.33523]
LCBR(-3)	0.003183	-0.296702
	(0.01149)	(0.10042)
	[0.27714]	[-2.95471]
LCBR(-4)	0.005213	0.087487
	(0.01162)	(0.10158)
	[0.44870]	[0.86127]
LCBR(-5)	-0.002987	-0.228348
	(0.01157)	(0.10112)
	[-0.25828]	[-2.25823]
LCBR(-6)	-0.007656	0.243779
	(0.01138)	(0.09950)
	[-0.67268]	[2.44993]
LCBR(-7)	0.004423	-0.345127
	(0.01143)	(0.09991)
	[0.38701]	[-3.45426]
LCBR(-8)	0.001847	0.218242
	(0.00794)	(0.06946)
	[0.23248]	[3.14181]
C	0.024311	0.233029
	(0.01199)	(0.10482)
	[2.02783]	[2.22315]
R-squared	0.999461	0.979971
Adj. R-squared	0.999413	0.978201
Sum-squared resid.	0.012482	0.954200
S.E. equation	0.008304	0.072607
F-statistic	20978.48	553.4977
Log likelihood	676.5487	247.2299
AIC	-6.662108	-2.325554
SIC	-6.379782	-2.043229

	LCCPI	LCBR
Mean dependent var.	4.441316	8.409428
SD dependent var.	0.342879	0.491766
Determinant resid. covariance (dof adj.)		3.63E-07
Determinant resid. covariance		3.03E-07
Log likelihood		924.0337
Akaike information criterion		-8.990239
Schwarz information criterion		-8.425587
Number of coefficients		34

Cointegrating Equation

Date: 10/15/20; time: 15:38

Sample (adjusted): 2003M10–2020M02

Included observations: 197 after adjustments

Trend assumption: linear deterministic trend

Series: LCCPI, LCBR

Lags interval (in first differences): 1–8

Unrestricted cointegration rank Test (trace)

Hypothesized		Trace	0.05	
No. of CE(s)	Eigenvalue	Statistic	Critical Value	Prob.**
None*	0.045996	15.70971	15.49471	0.0464
At most 1*	0.032129	6.433418	3.841465	0.0112

Notes: The trace test indicates two cointegrating equations at the 0.05 level.

*Denotes rejection of the hypothesis at the 0.05 level.

**MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted cointegration rank test (maximum eigenvalue)

Hypothesized		Max-eigen	0.05	
No. of CE(s)	Eigenvalue	Statistic	Critical Value	Prob.**
None	0.045996	9.276296	14.26460	0.2639
At most 1*	0.032129	6.433418	3.841465	0.0112

Notes: The maximum-eigenvalue test indicates no cointegration at the 0.05 level.

*Denotes rejection of the hypothesis at the 0.05 level.

**MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted cointegrating coefficients (normalized by $b_1 \times S11 \times b = 1$)

LCCPI	LCBR	
-6.705445	4.391282	
0.549704	1.964826	
Unrestricted adjustment coefficients (alpha)		
D(LCCPI)	0.000718	-0.001317
D(LCBR)	-0.013470	-0.005771
I Cointegrating equation(s)	Log likelihood	917.3583
Normalized cointegrating coefficients (standard error in parentheses)		
LCCPI	LCBR	
1.000000	-0.654883	
	(0.11762)	
Adjustment coefficients (standard error in parentheses)		
D(LCCPI)	-0.004817	
	(0.00403)	
D(LCBR)	0.090322	
	(0.03472)	

Error Correction Equation

Vector error correction estimates

Date: 10/15/20; time: 15:40

Sample (adjusted): 2003M10–2020M02

Included observations: 197 after adjustments

Note: Standard errors in () and t-statistic in [].

Cointegrating Eq.	CointEq1	
LCCPI(-1)	1.000000	
LCBR(-1)	-0.654883	
	(0.11762)	
	[-5.56763]	
C	1.066447	
Error Correction	D(LCCPI)	D(LCBR)
CointEq1	-0.004817	0.090322
	(0.00403)	(0.03472)
	[-1.19486]	[2.60128]
D(LCCPI(-1))	0.364207	1.198708
	(0.07451)	(0.64172)
	[4.88803]	[1.86796]
D(LCCPI(-2))	-0.108279	0.862300

Cointegrating Eq.	CointEqI	
	(0.07941)	(0.68393)
	[-1.36354]	[1.26081]
D(LCCPI(-3))	-0.094195	0.918518
	(0.07966)	(0.68610)
	[-1.18241]	[1.33875]
D(LCCPI(-4))	0.164897	-0.338199
	(0.08004)	(0.68936)
	[2.06016]	[-0.49060]
D(LCCPI(-5))	0.061674	-0.846612
	(0.07941)	(0.68397)
	[0.77660]	[-1.23780]
D(LCCPI(-6))	0.056650	0.544640
	(0.07970)	(0.68644)
	[0.71077]	[0.79343]
D(LCCPI(-7))	0.155948	-2.010207
	(0.07895)	(0.67997)
	[1.97526]	[-2.95634]
D(LCCPI(-8))	-0.026510	-0.030069
	(0.07724)	(0.66519)
	[-0.34324]	[-0.04520]
D(LCBR(-1))	0.006114	0.087187
	(0.00853)	(0.07349)
	[0.71647]	[1.18633]
D(LCBR(-2))	-0.004919	0.315301
	(0.00806)	(0.06942)
	[-0.61031]	[4.54205]
D(LCBR(-3))	-0.004129	0.022153
	(0.00850)	(0.07321)
	[-0.48571]	[0.30258]
D(LCBR(-4))	0.002892	0.103839
	(0.00834)	(0.07181)
	[0.34680]	[1.44601]
D(LCBR(-5))	-0.000580	-0.133298
	(0.00838)	(0.07221)
	[-0.06915]	[-1.84595]
D(LCBR(-6))	-0.005018	0.115655
	(0.00839)	(0.07227)

Cointegrating Eq.	CointEq1	
	[-0.59796]	[1.60024]
D(LCBR(-7))	-0.002421	-0.222723
	(0.00812)	(0.06994)
	[-0.29815]	[-3.18428]
D(LCBR(-8))	-0.008946	0.020669
	(0.00828)	(0.07132)
	[-1.08044]	[0.28983]
C	0.002711	0.002063
	(0.00099)	(0.00853)
	[2.73782]	[0.24187]
R-squared	0.230433	0.440980
Adj. R-squared	0.157346	0.387889
Sum-squared resid.	0.012747	0.945523
S.E. equation	0.008439	0.072679
F-statistic	3.152841	8.306076
Log likelihood	670.5669	246.3824
AIC	-6.625045	-2.318603
SIC	-6.325056	-2.018614
Mean dependent var.	0.006059	0.006297
SD dependent var.	0.009193	0.092895
Determinant resid. covariance (dof adj.)		3.75E-07
Determinant resid. covariance		3.09E-07
Log likelihood		917.3583
Akaike information criterion		-8.927496
Schwarz information criterion		-8.294187
Number of coefficients		38

LCCPI on LEX (Exchange Rate)

Vector autoregression estimates

Date: 10/15/20; time: 15:42

Sample (adjusted): 2003M03–2020M02

Included observations: 204 after adjustments

Note: Standard errors in () and t-statistic in [].

	LCCPI	LEX
LCCPI(-1)	1.331270	-0.108450
	(0.06620)	(0.07898)

	LCCPI	LEX
	[20.1087]	[-1.37318]
LCCPI(-2)	-0.337138 (0.06584)	0.112443 (0.07854)
	[-5.12063]	[1.43160]
LEX(-1)	0.032586 (0.05443)	1.394453 (0.06493)
	[0.59870]	[21.4763]
LEX(-2)	-0.028486 (0.05499)	-0.401114 (0.06560)
	[-0.51801]	[-6.11442]
C	0.009984 (0.02251)	0.016952 (0.02686)
	[0.44345]	[0.63116]
R-squared	0.999457	0.996899
Adj. R-squared	0.999446	0.996836
Sum-squared resid.	0.014328	0.020391
S.E. equation	0.008485	0.010123
F-statistic	91514.57	15991.86
Log likelihood	686.0272	650.0350
AIC	-6.676737	-6.323873
SIC	-6.595411	-6.242546
Mean dependent var.	4.419478	4.816927
SD dependent var.	0.360427	0.179970
Determinant resid. covariance (dof adj.)		7.35E-09
Determinant resid. covariance		6.99E-09
Log likelihood		1336.497
Akaike information criterion		-13.00487
Schwarz information criterion		-12.84222
Number of coefficients		10

Cointegration Equation

Date: 10/15/20; time: 15:43

Sample (adjusted): 2003M04–2020M02

Included observations: 203 after adjustments

Trend assumption: Linear deterministic trend

Series: LCCPI, LEX

Lags interval (in first differences): 1–2

Unrestricted cointegration rank test (trace)

Hypothesized		Trace	0.05	
No. of CE(s)	Eigenvalue	Statistic	Critical Value	Prob.**
None	0.045416	9.807236	15.49471	0.2959
At most 1	0.001830	0.371828	3.841465	0.5420

Notes: The trace test indicates no cointegration at the 0.05 level.

*Denotes rejection of the hypothesis at the 0.05 level.

**MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted cointegration rank test (maximum eigenvalue)

Hypothesized		Max-eigen	0.05	
No. of CE(s)	Eigenvalue	Statistic	Critical Value	Prob.**
None	0.045416	9.435408	14.26460	0.2517
At most 1	0.001830	0.371828	3.841465	0.5420

Notes: The maximum-eigenvalue test indicates no cointegration at the 0.05 level.

*Denotes rejection of the hypothesis at the 0.05 level.

**MacKinnon–Haug–Michelis (1999) p-values.

Unrestricted cointegrating coefficients (normalized by $b' \times S11 \times b = 1$)

LCCPI	LEX
–4.335158	3.653022
4.349376	–11.97259

Unrestricted adjustment coefficients (alpha)

D(LCCPI)	0.001728	9.81E-05
D(LEX)	–0.000477	0.000419
1 Cointegrating equation(s):	Log likelihood	1332.572

Normalized cointegrating coefficients (standard error in parentheses)

LCCPI	LEX
1.000000	–0.842650
	(0.44189)

Adjustment coefficients (standard error in parentheses)

D(LCCPI)	–0.007491
	(0.00255)
D(LEX)	0.002070
	(0.00310)

Appendix 3. Impulse Responses.

LCCPI on LCBR

Vector autoregression estimates

Date: 10/15/20; time: 15:05

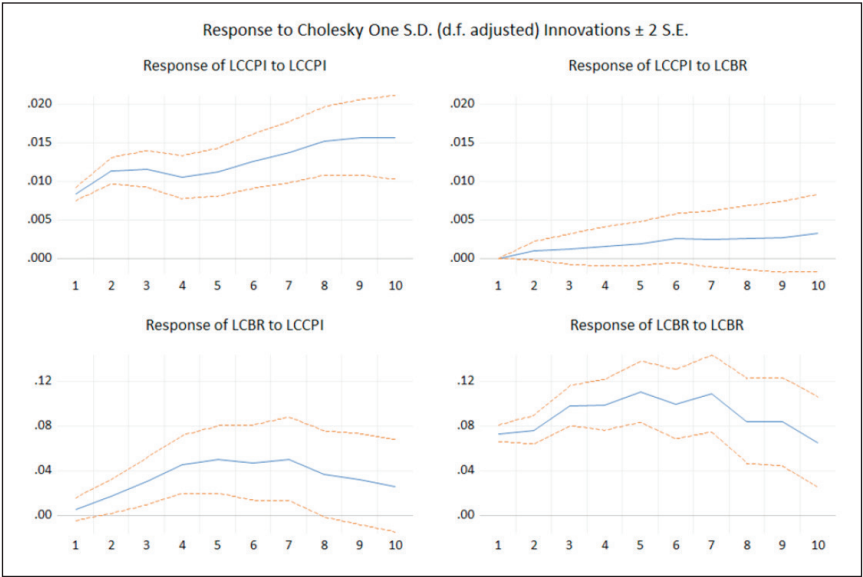
Sample (adjusted): 2003M09–2020M02

Included observations: 198 after adjustments

Note: Standard errors in () and t-statistic in [].

	LCCPI	LCBR
LCCPI(–1)	1.351748 (0.07352) [18.3866]	1.349171 (0.64422) [2.09426]
LCCPI(–2)	–0.472328 (0.12481) [–3.78435]	–0.483898 (1.09369) [–0.44244]
LCCPI(–3)	0.010073 (0.12982) [0.07759]	0.163576 (1.13758) [0.14379]
LCCPI(–4)	0.255388 (0.12960) [1.97064]	–1.230074 (1.13563) [–1.08317]
LCCPI(–5)	–0.090794 (0.12970) [–0.70005]	–0.545182 (1.13651) [–0.47970]
LCCPI(–6)	–0.005499 (0.12973) [–0.04239]	1.264456 (1.13677) [1.11233]
LCCPI(–7)	0.088179 (0.12491) [0.70594]	–2.456831 (1.09456) [–2.24459]
LCCPI(–8)	–0.143838 (0.07355) [–1.95562]	1.998198 (0.64451) [3.10034]
LCBR(–1)	0.012865 (0.00803) [1.60276]	1.045296 (0.07034) [14.8608]
LCBR(–2)	–0.014545 (0.01154) [–1.26062]	0.227765 (0.10111) [2.25272]

	LCCPI	LCBR
LCBR(-3)	0.002794 (0.01158) [0.24128]	-0.300427 (0.10148) [-2.96033]
LCBR(-4)	0.005096 (0.01172) [0.43495]	0.086367 (0.10267) [0.84120]
LCBR(-5)	-0.002865 (0.01166) [-0.24566]	-0.227181 (0.10221) [-2.22277]
LCBR(-6)	-0.007843 (0.01148) [-0.68333]	0.241985 (0.10057) [2.40606]
LCBR(-7)	0.004252 (0.01152) [0.36898]	-0.346758 (0.10099) [-3.43368]
LCBR(-8)	0.004306 (0.00792) [0.54379]	0.241814 (0.06939) [3.48487]
R-squared	0.999449	0.979424
Adj. R-squared	0.999403	0.977728
Sum-squared resid.	0.012766	0.980256
S.E. equation	0.008375	0.073390
F-statistic	22000.57	577.5574
Log likelihood	674.3247	244.5629
AIC	-6.649744	-2.308716
SIC	-6.384026	-2.042997
Mean dependent var.	4.441316	8.409428
SD dependent var.	0.342879	0.491766
Determinant resid. covariance (dof adj.)		3.76E-07
Determinant resid. covariance		3.17E-07
Log likelihood		919.4281
Akaike information criterion		-8.963920
Schwarz information criterion		-8.432483
Number of coefficients		32

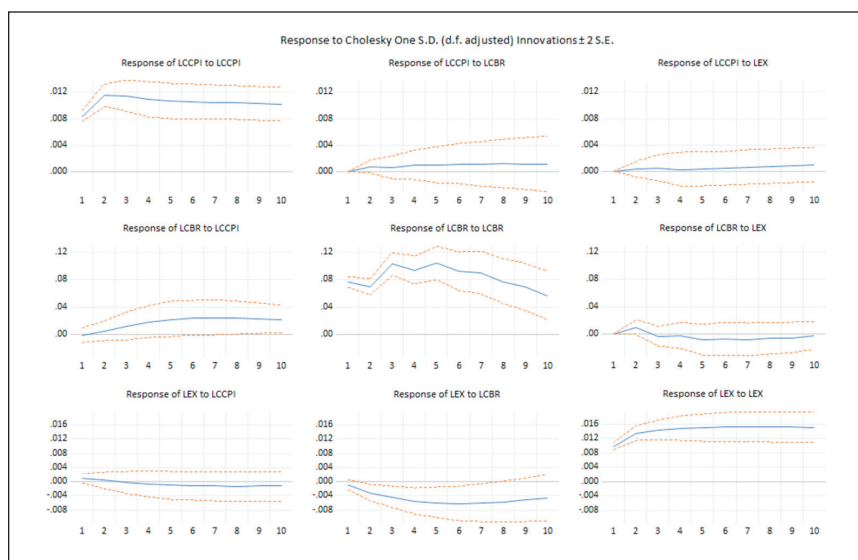


LCCPI on LCBR and LEX

Vector autoregression estimates
Date: 10/15/20; time: 15:09
Sample (adjusted): 2003M04–2020M02
Included observations: 203 after adjustments
Note: Standard errors in () and t-statistic in [].

	LCCPI	LCBR	LEX
LCCPI(−1)	1.372278 (0.07076) [19.3947]	0.664943 (0.64753) [1.02689]	−0.099795 (0.08422) [−1.18487]
LCCPI(−2)	−0.525049 (0.11624) [−4.51677]	0.322188 (1.06384) [0.30285]	0.100797 (0.13837) [0.72844]
LCCPI(−3)	0.143326 (0.07048) [2.03366]	−0.907868 (0.64499) [−1.40758]	−4.87E−05 (0.08389) [−0.00058]
LCBR(−1)	0.010616 (0.00662) [1.60346]	0.913995 (0.06059) [15.0849]	−0.025491 (0.00788) [−3.23448]

	LCCPI	LCBR	LEX
LCBR(-2)	-0.013667 (0.00889) [-1.53694]	0.520621 (0.08138) [6.39755]	0.018490 (0.01058) [1.74680]
LCBR(-3)	0.004301 (0.00674) [0.63818]	-0.517552 (0.06168) [-8.39124]	0.007680 (0.00802) [0.95738]
LEX(-1)	0.038351 (0.06074) [0.63136]	0.980417 (0.55591) [1.76363]	1.357645 (0.07231) [18.7761]
LEX(-2)	-0.064574 (0.09886) [-0.65319]	-2.633738 (0.90474) [-2.91105]	-0.371902 (0.11768) [-3.16031]
LEX(-3)	0.033647 (0.05896) [0.57067]	1.724416 (0.53959) [3.19576]	0.012792 (0.07018) [0.18226]
R-squared	0.999474	0.977741	0.997043
Adj. R-squared	0.999452	0.976823	0.996921
Sum-squared resid.	0.013600	1.139033	0.019270
S.E. equation	0.008373	0.076624	0.009967
F-statistic	46049.09	1065.176	8175.259
Log likelihood	687.4634	238.0326	652.0889
AIC	-6.684368	-2.256479	-6.335852
SIC	-6.537477	-2.109588	-6.188961
Mean dependent var.	4.423064	8.388490	4.818123
SD dependent var.	0.357650	0.503310	0.179600
Determinant resid. covariance (dof adj.)		4.02E-11	
Determinant resid. covariance		3.51E-11	
Log likelihood		1579.211	
Akaike information criterion		-15.29272	
Schwarz information criterion		-14.85205	
Number of coefficients		27	



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